SIRISHA GAJULA

F-1 OPT, STEM Eligible
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PROFESSIONAL SUMMARY

Data Engineer with 3 years of professional experience building ETL/ELT production pipelines, data validation frameworks, and reporting solutions on Microsoft Azure. Delivered 33 consecutive zero-failure production cycles processing 2M+ financial records monthly at Accenture. Graduate research at the University of New Haven extended this into ML-focused data engineering, including pipeline development for model consumption. Targeting roles at the intersection of data engineering and applied AI/ML. Microsoft Certified: Fabric Data Engineer Associate (DP-700). M.S. Data Science, University of New Haven (May 2026).

TECHNICAL SKILLS

Cloud & Data Integration: Azure Data Factory (ADF), Azure Data Lake Storage (ADLS), SSIS, Azure SQL Database, Azure Blob, Control-M, Azure Synapse Analytics (Foundational)

Data Engineering & Big Data: Azure Databricks, PySpark, Apache Spark, Spark SQL, Medallion Architecture (Bronze–Silver–Gold), ETL/ELT, Data Quality Validation

Databases & Programming: SQL Server, T-SQL (CTEs, Window Functions, Stored Procedures), Python, C#/.NET, pandas, NumPy

DevOps & Version Control: Azure DevOps, Git

Analytics & Machine Learning: Power BI, scikit-learn, LightGBM, FAISS, LLM APIs, Groq API

PROFESSIONAL EXPERIENCE

Accenture **Aug 2021 - Aug 2024**

Custom Software Engineering Analyst (Data Engineer) - Global Financial Reporting Platform *Hyderabad, India*

Azure Data Factory. Azure Synapse Analytics. SQL Server. SSIS. C# .NET. Control-M. Power BI. Azure DevOps. Git

- Developed SSIS ETL pipelines for new data sources and worked on migrating 5-6 legacy SSIS packages to Azure Data Factory (ADF) within a 20+ pipeline cloud migration project, transitioning ETL workflows from SharePoint and on-premises SQL Server to Azure Blob Storage and Azure SQL Database.
- Maintained and optimized SQL Server transformations using CTEs and window functions in SSMS, supporting **33 consecutive production cycles processing 2M+ financial records monthly** with zero reported data integrity failures.
- Performed C#-based source data validation by applying ETL transformation and formatting rules, updating validation logic for changing business requirements, and coordinating with QA teams to investigate data discrepancies.
- Supported migration of job scheduling from Control-M to Azure Data Factory, enabling automated pipeline scheduling, file validation, and record-count checks to reduce manual production monitoring and incident handling.
- Supported **Power BI dashboards and semantic models** for financial reporting across 3 business units, translating SQL business logic into KPI measures and communicating upstream schema and data changes with business stakeholders.

University of New Haven **Jan 2026 - May 2026**

Graduate Research Assistant, Data Engineering *West Haven, CT*

Python. PyTorch. MONAI. pandas . NumPy. NVIDIA A100. BraTS 2021

- Developed Python data pipelines for large-scale clinical imaging datasets (BraTS 2021, 1,251 cases), **cleaning, validating, and engineering features for ML model consumption**; applied data engineering practices to ensure model training inputs were accurate, complete, and consistently structured. Mean Dice 0.8625 outperforming MICCAI baseline.

PROJECTS

Financial RAG System Python. FAISS. MiniLM-L6-v2. Llama 3.1. Qwen3. Allam. Groq API. Streamlit GitHub

- Built a Retrieval-Augmented Generation pipeline for financial domain questions answering over a corpus of **~2,000+ Bloomberg/Reuters financial articles and central bank communications (~60MB)**, embedded with MiniLM-L6-v2 and indexed in FAISS. Compared three LLMs (Allam-2-7B, Llama-3.1-8B, Qwen3-32B) using **dual hallucination detection: cosine similarity between answer and retrieved evidence, and a semantic LLM judge** checking for unsupported content. Key finding: Llama-3.1-8B achieved the lowest hallucination rate despite not being the largest model. Deployed a Streamlit UI for interactive testing and side-by-side model comparison.

Sales Forecasting with LightGBM Python. LightGBM. XGBoost. Random Forest. pandas. scikit-learn. GridSearchCV GitHub

- Developed a sales forecasting system on historical e-commerce transaction data (1C Company dataset), aggregating daily records to monthly granularity. **Feature engineering included lag features, rolling statistics, temporal indicators, and price and demand signals** to capture seasonality and non-linear purchasing behavior. Benchmarked Random Forest, XGBoost, and LightGBM; LightGBM achieved the lowest RMSE, tuned via GridSearchCV across n_estimators, max_depth, learning_rate, subsample, and colsample_bytree.

Healthcare Data Engineering Pipeline Azure Databricks. PySpark. Delta Lake. Python. SQL. ADLS Gen2 GitHub

- Built a PySpark ETL pipeline in Azure Databricks using Bronze–Silver–Gold architecture, processing 3.9M+ healthcare records across 6 synthetic datasets. Implemented multi-run ingestion, schema handling, data transformations, and quality checks, developing Gold-layer tables for patient visits, costs, diagnosis trends, and provider analytics. Observed ~55% lower query execution time in a raw-versus-gold benchmark.

EDUCATION

Master of Science, Data Science - University of New Haven, West Haven, CT **May 2026**

Bachelor of Technology, Computer Science Engineering - Malla Reddy College of Engineering, India **May 2021**

CERTIFICATIONS

Microsoft Certified: Fabric Data Engineer Associate (DP-700). Earned June 2026. **Oracle: Agentic AI Foundations Associate (1Z0-1157-26)**. Earned July 2026. **NVIDIA: Building LLM Applications with Prompt Engineering**. **DataCamp Data Engineer Associate**